# About Me

- Ms. Marwa Khan
- Email-ID: Marwa.khan@nu.edu.pk
- Class Timings: 1:00pm – 4:00pm
- Grading: Absolute
- Attendance: 80%
- Google classroom Code: **clsarmp**

# Introduction to ICT

**Lecture 01**

# Course Outline & Marks Distribution

| Week | Topic |
|---|---|
| 1 | Introduction |
| 2 | Number System |
| 3 | Computer Organization |
| 4 | Mathematics in Computer Science |
| 5 | Operating systems |
| 6 | Data Management and its applications |
| 7 | Computer Graphics |
| 8 | Communication |
| 9 | Web development |
| 10 | Artificial Intelligence |
| 11 | Big data |
| 12, 13, 14 | Student Presentations |
| | FINAL EXAM |

| | |
|---|---|
| • **Class Participation, Assignments & Quizzes** | **25%** |
| • **Final Presentation** | **25%** |
| • **Exam** | **50%** |

# Introduction to Information and Communication Technology (IICT)

- What is IICT?
- Significance of IICT
  - Why we need to study this course?
- Looking into the impact of Computers in our lives.

# ICT

- Information
    - Any entity or form that provides the answer to a Query
- Communication
    - Act of transferring information from one place to another
- Technology
    - Collection of techniques, methods and processes used to produce goods or services or accomplish objectives.

# The Meaning of ICT

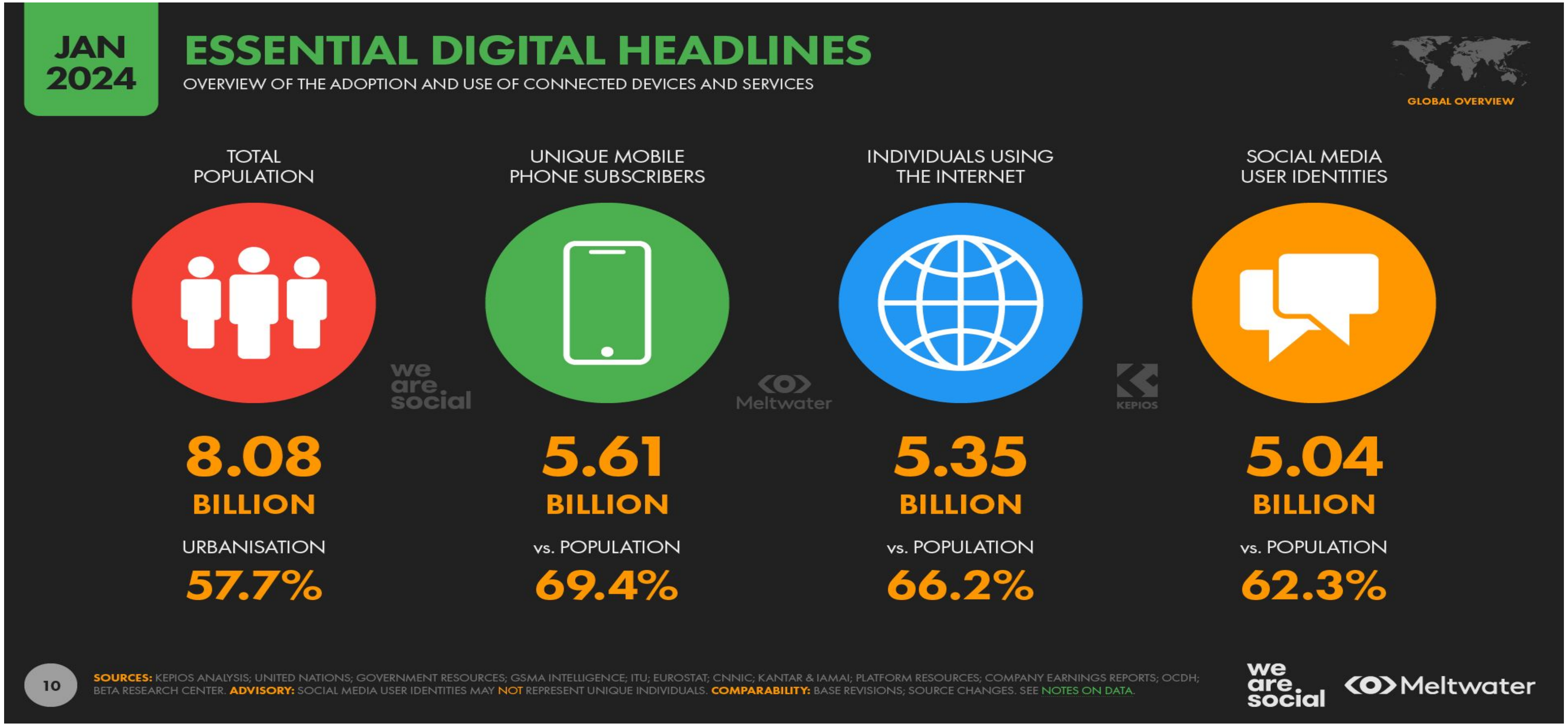
- **I**nformation
- **C**ommunication
- **T**echnologies

**ICT** are the hardware and software that enable society to create, collect, consolidate and communicate information in different formats and for various purposes.

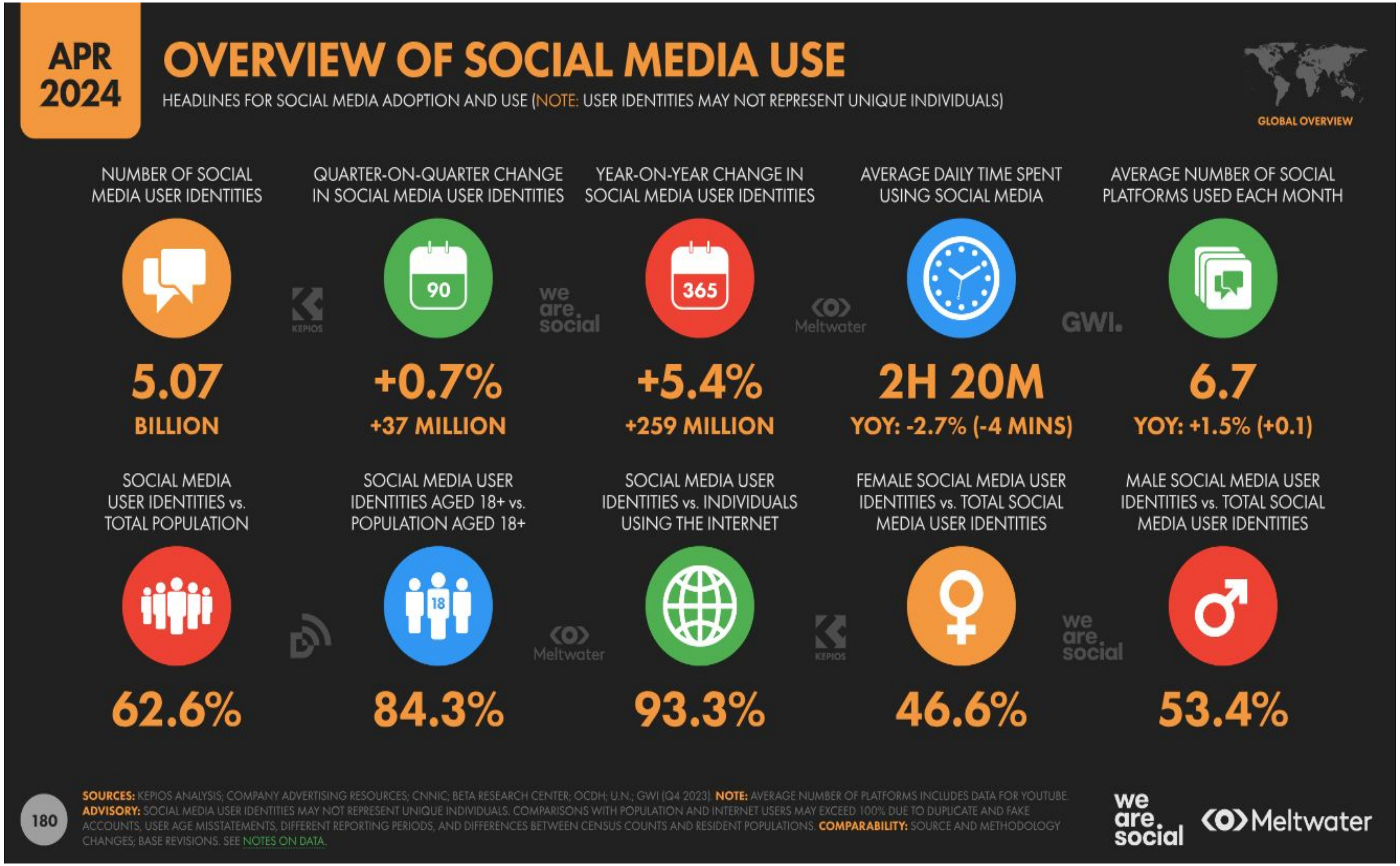
Do you use an ICT


ATMcard
ANGRY BIRDS
UBER
TAXI
ONLINE SHOPPING
View my items
Checkout
The Slanted Door
ANGRY BIRDS
Podcasts
Photos
Camera
Evernote
Messages
WhatsApp
Messenger
Instagram
ATMcard

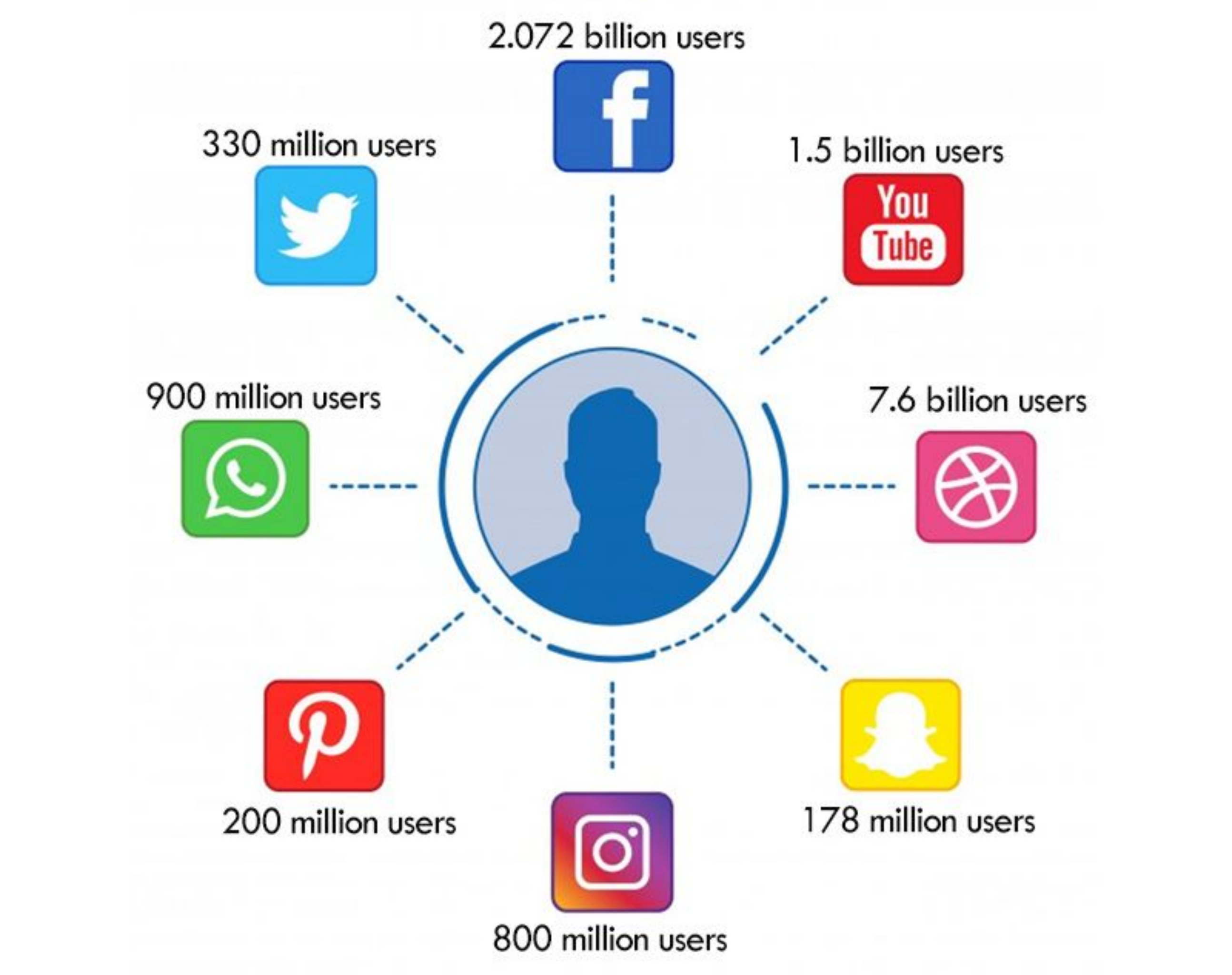
# Amazing Facts and Statistics



# Amazing Facts and Statistics

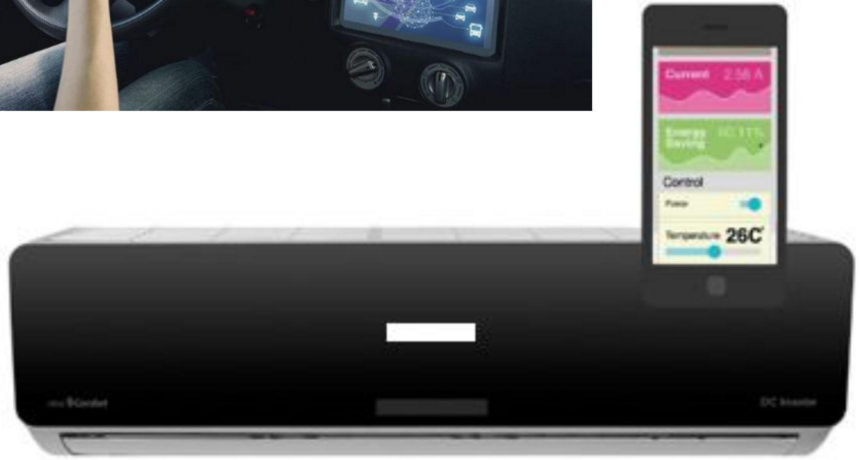


# Amazing Facts and Statistics



# Cont…

- Computers are penetrating into every aspects of our lives
    - Air Conditioners
    - Cars
    - Airplanes
    - Smartphones
    - Etc…


Some examples of Computers (Embedded Systems) in our daily life

So, these computers are taking over our lives.

Is this good or bad?

# Bright side of Computer

- Enable new discoveries
- Lead to efficiencies
- Making Life easy

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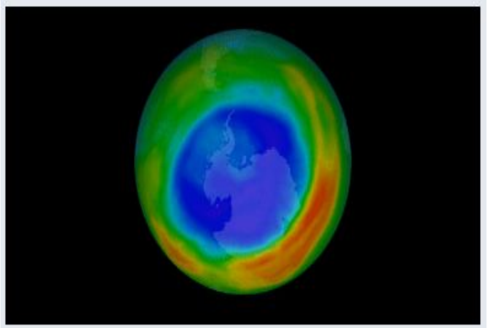
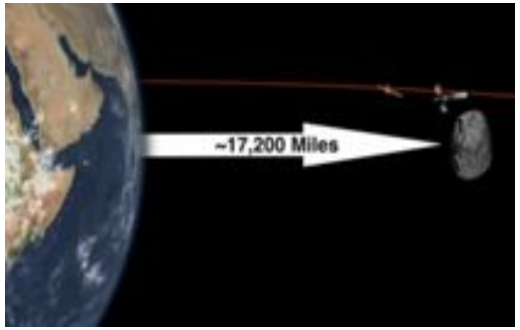


NASA Sees First Direct Proof of Ozone Hole Recovery



# Making Life Easy

Today, computers are used in every/most of the aspects of our lives.

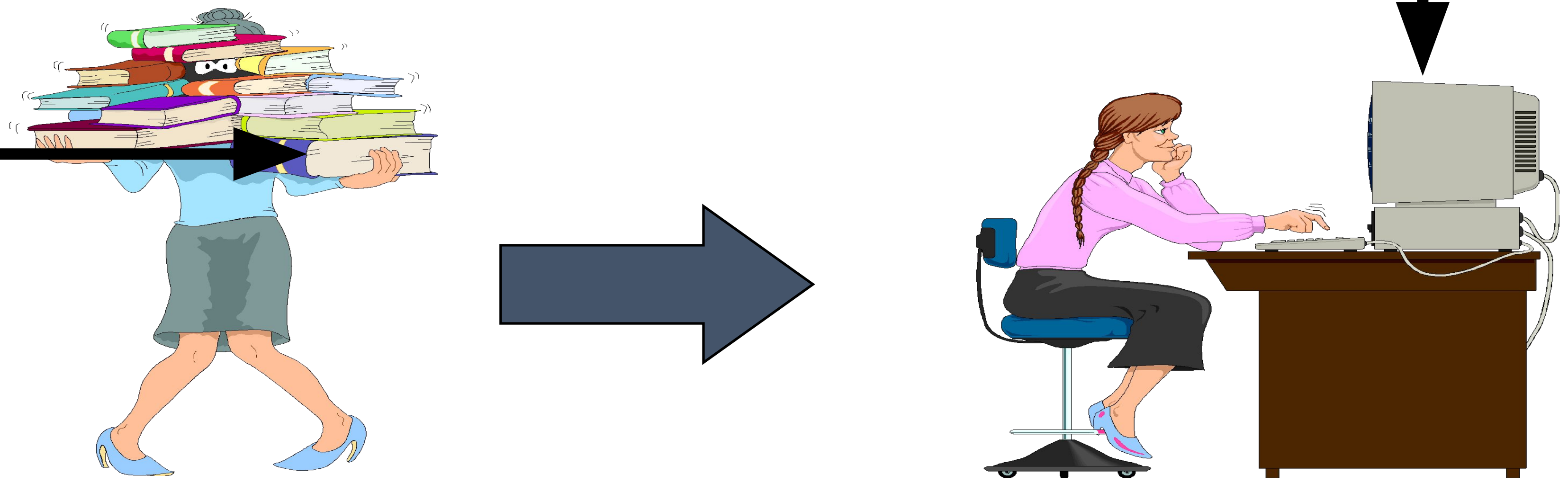
Some aspects are given below:

- Social Media/Communication
- E-Commerce/ M-commerce
- Automation
- Defense
- Education and Science & Technology
- Recreational Aspects

# Transition

## Shift from Print to Digital

Internet



**Innovations in ICT have made the transfer of digital information from remote sites possible**

# Social Media/Communication

Some examples of Social Media

- Facebook
- Twitter
- LinkedIn
- Facebook Messenger
- WhatsApp
- …

# E-Commerce

- Amazon
- Daraz.pk
- UBER
- Air/Bus ticket booking and seat reservation

# Automation

- Home automation system
  - Example: JARVIS by FACEBOOK
- Car assembly automation
- Food products packing

# Defense/Military

- Missile Control
- Military Communication
- Military Operation and Planning
- Smart Weapons

# Education

- Education
  - The computer provides a tool in the education system known as CBE (Computer Based Education).
  - CBE involves control, delivery, and evaluation of learning.
  - Computer education is rapidly increasing the graph of number of computer students.
  - There are a number of methods in which educational institutions can use a computer to educate the students.
  - It is used to prepare a database about performance of a student and analysis is carried out on this basis.

# Recreational and Entertainment Aspect

- Some examples?
- Animated Movies
- High graphics games

# Basics

Computer

Hardware

Software

CPU

Memory

I/O

Application Software

System Software

# Hardware: Processor

- Processor consists of 2 components: ALU & CU
- The control unit (CU) of the CPU contains circuitry that uses electrical signals to direct the entire computer system to carry out, or execute, stored program instructions.
- The arithmetic/logic unit (ALU) contains the electronic circuitry that executes all arithmetic and logical operations.

# Hardware: Memory

- Primary Memory
  - RAM, SRAM, DRAM, SDRAM
  - ROM, EROM, EPROM, EEPROM
- Secondary Memory
  - CD, DVD, Memory Cards
  - SD Cards, Flash Drives

# Hardware & Software

| Hardware | Software |
|---|---|
| Physical elements of a computer or electronic system | A collection of instructions that tells the computer how to perform a task |
| Categories: I/O devices, Processor, Memory | Mainly divided into system software and application software |
| Tangible | Intangible |
| Developed using electronic and other materials | Developed by writing instructions using programming languages |
| When damages, it can be replaced with a new component | When damages, it can be reinstalled using a backup copy |
| Starts functioning once the software is loaded into the system | Should be installed into the computer to function |
| Examples: Keyboard, Mouse, CPU, ROM, RAM | Examples: MS Word, Excel, Whatsapp, Instagram |

# System Vs Application (Software)

| System Software | Application Software |
| --- | --- |
| System software controls hardware. | Application software fulfils user requests |
| Acts as an interface between hardware and application software | Runs on the platform provided by system software |
| Examples include, Operating Systems, device drivers, etc. | Word processors, media players, etc. |

# Types of Computer

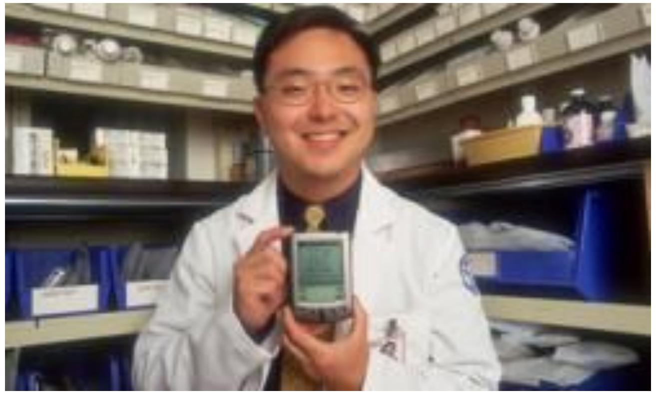
- PC (Personal Computer )
  - Small
  - Inexpensive
  - Business Use
    - Word processing, running spreadsheets etc.
  - Personal Use
    - Playing games, surfing the internet, movies, games etc.
  - Single user system

# Types of Computer

- Desktop
  - PC not designed for portability
  - More storage and computation with less cost
- Laptop (notebook)
  - Portable with integrated display
  - Battery operated

# Types of Computer

- Netbook/Notebook
  - Smaller and cheaper than laptops
  - Less powerful than laptop
- PDA (personal digital assistant)
  - Tightly integrated computer
  - Flash memory instead of hard disk
  - Touch screen instead of keyboard
  - Lightweight and reasonable battery life



# Types of Computer

- Workstation
  - Desktop computer with more processing power
  - More memory
  - More capabilities in performing specialized tasks

# Types of Computer

- Server
  - Computer that serves other computers over network
  - More processing power, memory and storage
  - Large in size

# Types of Computer

- Mainframe
  - Very large size
  - Now known as enterprise server
  - More processing power

- Supercomputer
  - Very expensive
  - Fastest computers
  - Employed for specific applications which require immense amount of calculations
    - weather forecasting
    - scientific simulations
    - (animated) graphics
    - nuclear energy research
    - electronic design

# Advantages of Computer

- High Speed
  - Units of speed in microsecond and nanosecond
  - It can perform millions of calculations in few seconds where it will take months for a human to perform same calculations
- Accuracy
  - 100 % error free
- Storage
  - More storage capacity than humans
  - All types of images can be stored

# Advantages of Computer

- Diligence
    - Free from tiredness and lack of concentration
    - Can work continuously without error and boredom
    - Repeated task with same time and accuracy
- Versatility
    - solves different problems
        - Solving complex problems
        - Games
- Reliability
    - Machines with long lives
    - Easy maintenance

# Advantages of Computer

- Automation
  - Computer is automatic machine
  - After taking input and program, then it can process without human interaction
- Reduction in manual work and paper cost
  - Data stored inside computers electronically reduces paper cost
  - Increases speed
  - Data retrieved easily and speedily when required

# Weaknesses of Computer

- No I.Q.
  - No intelligence
  - Instructions given to computer
  - Cannot take decisions
- Dependency
  - Depends on instructions given by humans
- No Feeling
  - Cannot make judgment on feelings, taste, experience, knowledge

# Cons

- High risk of privacy invasion
- Health implications
- Social Disconnect
- Addiction

**Trump to Kim: My nuclear button is 'bigger and more powerful'**

Could Donald Trump press the button?

Mr Trump's latest comment states the obvious: a US president has immediate access to the nuclear codes.

However, **the real process of launching a nuclear attack** does not involve any button-pressing.

After high-level consultations, the president would exchange codes with top military officials. They are printed on a card known as "the biscuit", which he carries wherever he goes.

http://www.bbc.com/news/world-asia-42549687

But we have a choice:
What to do about/with this beast (Computer)?

# End of The Lecture